

CC LAB 8

NAME: AVANEE UPADHYAYA

SRN: PES2UG23CS108

SECTION: B

SS1:

```
avaneer@Avaneer:~/Desktop/raft_docker_lab$ cd raft_docker_lab
ls
docker-compose.yml  Dockerfile  node.py  requirements.txt
avaneer@Avaneer:~/Desktop/raft_docker_lab/raft_docker_lab$
```

SS2A:

```
[+] up 7/7
✓Image raft_docker_lab-node2      Built
✓Image raft_docker_lab-node3      Built
✓Image raft_docker_lab-node1      Built
✓Network raft_docker_lab-raftnet  Created
✓Container node3                  Created
✓Container node1                  Created
✓Container node2                  Created
Attaching to node1, node2, node3
node3 | Starting Node 3 at 172.28.0.4:8003
node3 | [Node 3] Waiting for leader election...
node1 | Starting Node 1 at 172.28.0.2:8001
node1 | [Node 1] Waiting for leader election...
node2 | Starting Node 2 at 172.28.0.3:8002
node2 | [Node 2] Waiting for leader election...
node3 | [Node 3] Leader changed: 172.28.0.4:8003
node3 | [LEADER Node 3] I am the leader
node1 | [Node 1] Leader changed: 172.28.0.4:8003
```

```
node1 | [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node3 | [LEADER Node 3] I am the leader
```

[illegible]

SS3C:

```
avanee@Avanee: ~/Desktop/ | X + v
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

SS3D:

```
avanee@Avanee: ~/Desktop/ | X + v
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

SS4:

```
avanee@Avanee:~$ cd ~/Desktop/raft_docker_lab/raft_docker_lab
docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS        NAMES
ba5d5c0336f8   raft_docker_lab-node3 "python node.py 3"      15 minutes ago Up 15 minutes        node3
7e2310b1e337   raft_docker_lab-node2 "python node.py 2"      15 minutes ago Up 15 minutes        node2
79675a82462e   raft_docker_lab-node1 "python node.py 1"      15 minutes ago Up 15 minutes        node1
avanee@Avanee:~/Desktop/raft_docker_lab/raft_docker_lab$
```

SS5:

```
avane@Avane:~/Desktop/raft_docker_lab/raft_docker_lab$ docker network inspect raft_docker_lab_raftnet
[
  {
    "Name": "raft_docker_lab_raftnet",
    "Id": "4b4c71237a78b2a983671364129bf901e5580f86f07b7f7e332c6521053322d8",
    "Created": "2026-03-26T08:52:52.938601594Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.28.0.0/16",
          "Gateway": "172.28.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {
      "com.docker.compose.config-hash": "7b955128463677559d4edba13e9fe8f69dd803b2dfc190dd0f2333946fc6a5d0",
      "com.docker.compose.network": "raftnet",
      "com.docker.compose.project": "raft_docker_lab",
      "com.docker.compose.version": "5.0.2"
    },
    "Containers": {
      "79675a82462e4724773695191a2467faa259025597c0c19455df9f4fb4cfb475": {
        "Name": "node1",
        "EndpointID": "205164e296fe1aa2ef4811746b5d31e49755d161531783f0d9059a25d302b09b",
        "MacAddress": "d6:07:97:98:77:52",
        "IPv4Address": "172.28.0.2/16",
        "IPv6Address": ""
      }
    }
  }
]
```

```
    "com.docker.compose.version": "5.0.2"
  },
  "Containers": {
    "79675a82462e4724773695191a2467faa259025597c0c19455df9f4fb4cfb475": {
      "Name": "node1",
      "EndpointID": "205164e296fe1aa2ef4811746b5d31e49755d161531783f0d9059a25d302b09b",
      "MacAddress": "d6:07:97:98:77:52",
      "IPv4Address": "172.28.0.2/16",
      "IPv6Address": ""
    },
    "7e2310b1e337bca1bcc4882c79fdff13ca246f92c32056b148434b5cef3f168": {
      "Name": "node2",
      "EndpointID": "e6967765238c97a3aa0677d3d0d6da1bc744449a15c2ec372bcfea5d68cb11a3",
      "MacAddress": "2e:90:16:94:21:9d",
      "IPv4Address": "172.28.0.3/16",
      "IPv6Address": ""
    },
    "ba5d5c0336f8bbcb246bdf1624917d9e3370b540ebfa17c50bdc4ac0899e13e": {
      "Name": "node3",
      "EndpointID": "a2a2f38b8a71c92caaa5842bdbaa8e6a20a4816817dd4d26a5767b23c4ad3711",
      "MacAddress": "ee:9a:03:f6:3b:04",
      "IPv4Address": "172.28.0.4/16",
      "IPv6Address": ""
    }
  },
  "Status": {
    "IPAM": {
      "Subnets": {
        "172.28.0.0/16": {
          "IPsInUse": 6,
          "DynamicIPsAvailable": 65530
        }
      }
    }
  }
}
```

SS6A:

```
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node3 exited with code 137
node1 | [Node 1] Leader changed: None
node1 | [Node 1] Waiting for leader election...
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node2 | [Node 2] Leader changed: None
node2 | [Node 2] Waiting for leader election...
node1 | [Node 1] Waiting for leader election...

node2 | [Node 2] Waiting for leader election...
node1 | [Node 1] Waiting for leader election...

node1 | [Node 1] Waiting for leader election...
node2 | [Node 2] Waiting for leader election...

node1 | [Node 1] Leader changed: 172.28.0.3:8002
node2 | [Node 2] Leader changed: 172.28.0.3:8002
node2 | [LEADER Node 2] I am the leader
node1 | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

SS6B:

✕ + ▾

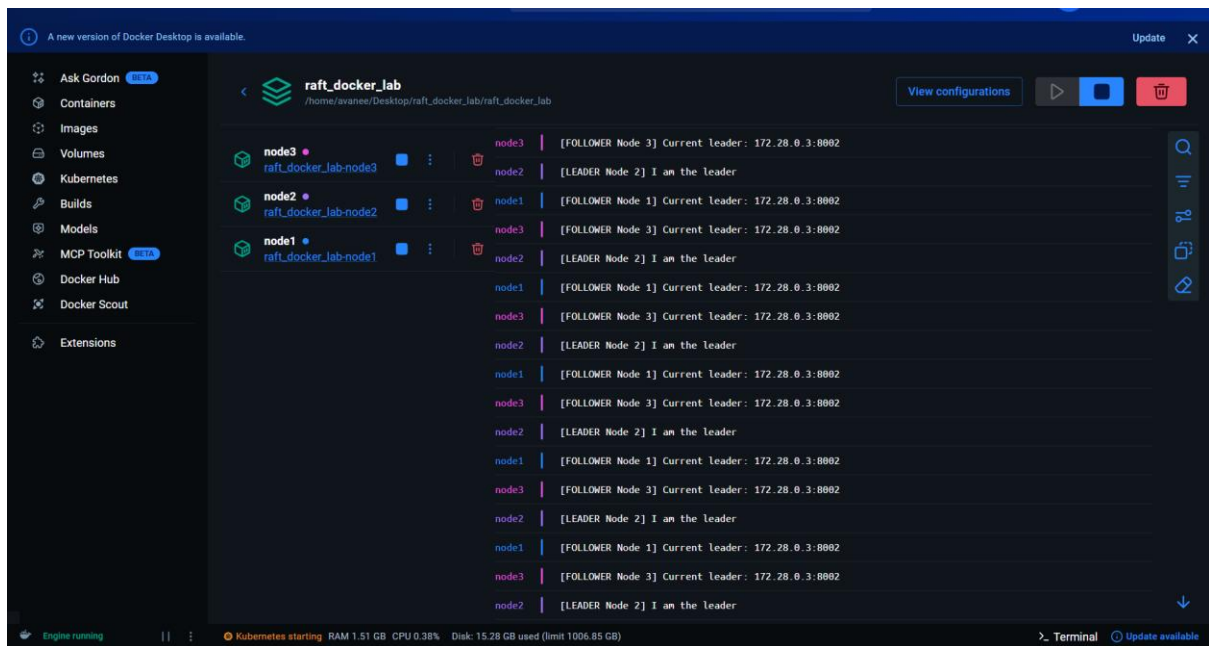
[illegible]

SS6C:

✕ + ▾

[illegible]

SS7A:



SS7B:

```
avaneer@Avaneer:~$ docker start node3
node3
avaneer@Avaneer:~$ docker logs --tail 20 node3
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
avaneer@Avaneer:~$
```

SS8:

```

avanee@Avanee: ~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
7e28737311e6   raft_docker_lab-node2              "python node.py 2"     18 seconds ago Up 17 seconds node2
445523b96938   raft_docker_lab-node3              "python node.py 3"     18 seconds ago Up 17 seconds node3
697abfc6b6f1a   raft_docker_lab-node5              "python node.py 5"     18 seconds ago Up 17 seconds node5
7c1e8f52eb77   raft_docker_lab-node4              "python node.py 4"     18 seconds ago Up 17 seconds node4
0e50e195e69a   raft_docker_lab-node1              "python node.py 1"     18 seconds ago Up 17 seconds node1
avanee@Avanee: ~$

```

SS9:

[illegible]

SS10:

[illegible]

SS11:

[illegible]

SS12:

[illegible]

SS13:

[illegible]

SS14:

```

avaneer@Avaneer: ~
avaneer@Avaneer:~$ docker stop node4
docker stop node5
node4
node5
avaneer@Avaneer:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
7e28737311e6   raft_docker_lab-node2              "python node.py 2"      8 minutes ago Up 8 minutes   node2
445523b96938   raft_docker_lab-node3              "python node.py 3"      8 minutes ago Up 8 minutes   node3
0e50e195e69a   raft_docker_lab-node1              "python node.py 1"      8 minutes ago Up 8 minutes   node1
avaneer@Avaneer:~$

```

SS15:

```

avanee@Avanee: ~
avanee@Avanee:~$ docker stop node3
node3
avanee@Avanee:~$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS        NAMES
7e28737311e6   raft_docker_lab-node2  "python node.py 2"      9 minutes ago Up 9 minutes        node2
0e50e195e69a   raft_docker_lab-node1  "python node.py 1"      9 minutes ago Up 9 minutes        node1
avanee@Avanee:~$

```

SS16:

[illegible]

```
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[Node 1] Leader changed: None
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
```

SS17:

[illegible]